Joshua Kazdan, Rylan Schaeffer, Apratim Dey, Matthias Gerstgrasser, Rafael Rafailov, David L. Donoho, and Sanmi Koyejo. 2025. Collapse or thrive? perils and promises of synthetic data in a self-generating world. *Preprint*, arXiv:2410.16713.

Ying Li, Zhidi Lin, Feng Yin, and Michael Minyi Zhang. 2024. Preventing model collapse in gaussian process latent variable models. *Preprint*, arXiv:2404.01697.

Keith Man and Javaan Chahl. 2022. A review of synthetic image data and its use in computer vision. *Journal of Imaging*, 8(11).

Gonzalo Martínez, Lauren Watson, Pedro Reviriego, José Alberto Hernández, Marc Juarez, and Rik Sarkar. 2023. Combining generative artificial intelligence (ai) and the internet: Heading towards evolution or degradation? *Preprint*, arXiv:2303.01255.

Zach Nussbaum, John X. Morris, Brandon Duderstadt, and Andriy Mulyar. 2024. Nomic embed: Training a reproducible long context text embedder. *Preprint*, arXiv:2402.01613.

Jorge Pérez, Javier Marinković, and Pablo Barceló. 2019. On the turing completeness of modern neural network architectures. *Preprint*, arXiv:1901.03429.

Rylan Schaeffer, Joshua Kazdan, Alvan Caleb Arulandu, and Sanmi Koyejo. 2025. Position: Model collapse does not mean what you think. *Preprint*, arXiv:2503.03150.

Mohamed El Amine Seddik, Suei-Wen Chen, Soufiane Hayou, Pierre Youssef, and Merouane Debbah. 2024. How bad is training on synthetic data? a statistical analysis of language model collapse. *Preprint*, arXiv:2404.05090.

Ilia Shumailov, Zakhar Shumaylov, Yiren Zhao, Yarin Gal, Nicolas Papernot, and Ross Anderson. 2024. The curse of recursion: Training on generated data makes models forget. *Preprint*, arXiv:2305.17493.

Aditi Singh, Abul Ehtesham, Saket Kumar, and Tala Talaei Khoei. 2025. Agentic retrieval-augmented generation: A survey on agentic rag. *Preprint*, arXiv:2501.09136.

Ananda Theertha Suresh, Andrew Thangaraj, and Aditya Nanda Kishore Khandavally. 2024. Rate of model collapse in recursive training. *Preprint*, arXiv:2412.17646.

Pablo Villalobos, Anson Ho, Jaime Sevilla, Tamay Besiroglu, Lennart Heim, and Marius Hobbhahn. 2024. Will we run out of data? limits of llm scaling based on human-generated data. *Preprint*, arXiv:2211.04325.

Edward L. Wang, Tianyu Wang, Avanti Athreya, Vince Lyzinski, and Carey E. Priebe. 2025. Gaussian mixture models as a proxy for interacting language models. *Preprint*, arXiv:2506.00077.

Guan Wang, Sijie Cheng, Xianyuan Zhan, Xiangang Li, Sen Song, and Yang Liu. 2023. Openchat: Advancing open-source language models with mixed-quality data. *arXiv preprint arXiv:2309.11235*.

Sierra Wyllie, Ilia Shumailov, and Nicolas Papernot. 2024. Fairness feedback loops: Training on synthetic data amplifies bias. *Preprint*, arXiv:2403.07857.

Greg Yang. 2020. Tensor programs ii: Neural tangent kernel for any architecture. *Preprint*, arXiv:2006.14548.

Zoran Zivkovic and Ferdinand van der Heijden. 2004. Recursive unsupervised learning of finite mixture models. *IEEE Transactions on pattern analysis and machine intelligence*, 26(5):651–656.

A Sketch of Proof for Conjecture (5)

In this section, we discuss our conjecture (5) mathematically. In the GMM experiment, we measure the GMMs' weight differences in the distance matrix, so we start with proving pairwise weight differences going to 0.

For now, suppose we do not do the positive normalization step in Algorithm 2. Let us consider the update of GMM $i = 1, \ldots, N$ with points $u_1^{(t)}, \ldots, u_{k_t}^{(t)}$ sampled uniformly at random without replacement from $A^{(t)}$ (we denote this distribution as $\mathcal{U}(A^{(t)})$). Let us also consider the update for just one $u_j^{(t)}$ (i.e., one iteration of the outer for-loop in Algorithm 2). The mixture-weight update for component $b = 1, \ldots, B$ is then

$$\pi_{i,b}^{(t+1)} \leftarrow (1-\alpha_t)\pi_{i,b}^{(t)} + \frac{\alpha_t}{1 - Bc_{t,B}} \left\{ o_{i,b}^{(t)}(u_j^{(t)}) - c_B \right\}. \tag{6}$$

Let us integrate the RHS over $u_j^{(t)} \sim \mathcal{U}(A^{(t)})$ for some arbitrary $j \in \{1, \ldots, k_t\}$. For mathematical simplicity, suppose $A^{(0)}$ is empty. Under this simplifying assumption, the set $A^{(t)}$ contains only points generated from a (possibly previous) GMM, so the update point $u_j^{(t)}$ must have been generated by some GMM in an earlier step. Because the k_t update points $u_1^{(t)}, u_2^{(t)}, \ldots, u_{k_t}^{(t)}$ are drawn simultaneously from the same distribution, the order in which an update point is drawn does not affect the probability of which GMM (among $\{(i', t') \colon i' \in [n], t' \in [t-1]\}$) the point is drawn from. Hence, we can say that $u_j^{(t)}$ is drawn from GMM $g_{i'}^{(t')}$ with probability $p_{i'}^{(t')} = (nt)^{-1}$. We then integrate the ownership $o_{i,b}^{(t)}(u_j^{(t)})$, which is

the only term in the RHS of (6) that involves the update point:

$$\int o_{i,b}^{(t)}(u)\,\mathrm{d}g_{i'}^{(t')}(u) \equiv \int \frac{\pi_{i,b}^{(t)} N(u;\mu_b,\Sigma)}{g_i^{(t)}(u)} \mathrm{d}g_{i'}^{(t')}(u) \tag{7}$$

$$= \pi_{i,b}^{(t)} \int \frac{g_{i'}^{(t')}(u)}{g_i^{(t)}(u)} \mathrm{d}N(u;\mu_b,\Sigma) \tag{8}$$

where $g_i^{(t)}(u) = \sum_{\ell=1}^{B} \pi_{i,\ell}^{(t)} N(u;\mu_b,\Sigma)$ is the likelihood of GMM i at u. Because the GMM component parameters are fixed (only the mixture weights vary), if the mixture variances are small enough (almost same as saying if the mixture components are well-separated enough), then we can make the approximation $\frac{g_{i'}^{(t')}(u)}{g_i^{(t)}(u)} \approx \sum_{\ell=1}^{B} \frac{\pi_{i',\ell}^{(t')}}{\pi_{i,\ell}^{(t)}} 1_{(u=\mu_\ell)}$. We also have that $N(u;\mu_b,\Sigma)$ is almost zero if u is far enough away from μ_b, so we can approximate

$$\int \frac{g_{i'}^{(t')}(u)}{g_i^{(t)}(u)} N(u;\mu_b,\Sigma)\mathrm{d}u \approx \frac{\pi_{i',b}^{(t')}}{\pi_{i,b}^{(t)}}$$

so then Eq.(7) is approximated as $\pi_{i',b}^{(t')}$. Under this assumption, the integral of the RHS of Eq.(6) over $u_j^{(t)} \sim \mathcal{U}(A^{(t)})$ is approximately

$$\approx (1-\alpha_t)\pi_{i,b}^{(t)} + \alpha_t \frac{\bar{\pi}_b^{(t)} - c_{t,B}}{1 - Bc_{t,B}} \tag{9}$$

where we introduce the notation

$$\bar{\pi}_b^{(t)} := \sum_{i'=1}^{n}\sum_{t'=0}^{t-1} p_{i'}^{(t')}\pi_{i',b}^{(t')} = (nt)^{-1}\sum_{i'=1}^{n}\sum_{t'=0}^{t-1}\pi_{i',b}^{(t')},$$

which is component b's mixture weight averaged over all n models and all past times $0,\ldots,t-1$. If $0 < c_{t,B} < \min\{B^{-1}, \bar{\pi}_b^{(t)}\}$, then $\frac{\bar{\pi}_b^{(t)} - c_{t,B}}{1-Bc_{t,B}}$ in (9) is positive, which guarantees that the entire term in (9) is also positive.

If we do this update for each of the k_t points, the ultimate update (in expectation) for $\pi_{i,b}^{(t+1)}$ is

$$\approx \bar{\alpha}_t^{k_t}\pi_{i,b}^{(t)} + \alpha_t \frac{\bar{\pi}_b^{(t)} - c_{t,B}}{1 - Bc_{t,B}} \sum_{s=0}^{k_t-1} \bar{\alpha}_t^s$$

$$= \bar{\alpha}_t^{k_t}\pi_{i,b}^{(t)} + \{1 - \bar{\alpha}_t^{k_t+1}\}\left[\frac{\bar{\pi}_b^{(t)} - c_{t,B}}{1 - Bc_{t,B}}\right]$$

where we introduce the notation $\bar{\alpha}_t = 1 - \alpha_t$. Thus, at each t, the updated mixture-weight $\pi_{i,b}^{(t+1)}$ will, in expectation, be a convex combination of the (scaled) mixture weight $\pi_{i,b}^{(t)}/(1-\alpha_t)$ and the value in the square brackets, which does not depend on which of the n GMMs is being updated. In particular, for any i and i' where $i \neq i'$, we have

$$E_t\left[\pi_{i,b}^{(t+1)} - \pi_{i',b}^{(t+1)}\right] \approx (1-\alpha_t)^{k_t}\left[\pi_{i,b}^{(t)} - \pi_{i',b}^{(t)}\right].$$

where E_t denotes the conditional expectation w.r.t. the distribution described in Algorithms 2 and 3 given all quantities from all times up to and including time t. By iterating over conditional expectations, we get

$$E_0\left[\pi_{i,b}^{(t+1)} - \pi_{i',b}^{(t+1)}\right] \approx \left[\prod_{s=0}^{t}(1-\alpha_s)^{k_s}\right]\left[\pi_{i,b}^{(0)} - \pi_{i',b}^{(0)}\right].$$

If $\alpha_t = k_t^{-1}$, which is suggested by Zivkovic and van der Heijden (2004), then because $\lim_{k_t\to\infty}(1-\alpha_t)^{k_t} = e^{-1}$ is a nonnegative constant strictly smaller than 1 and $\lim_{t\to\infty} k_t = \infty$, we conclude that $\lim_{t\to\infty} E_0[\pi_{i,b}^{(t)} - \pi_{i',b}^{(t)}] = 0$ and thus also

$$\lim_{t\to\infty} E_0|\pi_{i,b}^{(t)} - \pi_{i',b}^{(t)}| = 0. \tag{10}$$

Note that

$$\begin{aligned} E_0\left[\|D^{(t)}\|_F^2\right] &= E_0\left[\sum_{i,i'}\left|D_{i,i'}^{(t)}\right|^2\right] \\ &= E_0\left[\sum_{i,i'}^{n}\sum_{b=1}^{B}\left|\pi_{i,b}^{(t)} - \pi_{i',b}^{(t)}\right|^2\right] \\ &= \sum_{i,i'}^{n}\sum_{b=1}^{B} E_0\left[\left|\pi_{i,b}^{(t)} - \pi_{i',b}^{(t)}\right|^2\right] \\ &\leq \sum_{i,i'}^{n}\sum_{b=1}^{B} E_0\left|\pi_{i,b}^{(t)} - \pi_{i',b}^{(t)}\right| \end{aligned}$$

where the inequality comes from the fact that each mixture weight is a value between 0 and 1. Thus, (10) implies

$$\lim_{t\to\infty} E_0\left[\|D^{(t)}\|_F^2\right] = 0.$$

Because $E[X^2] \geq E^2[X]$ for any random variable X and expectation E, we also get

$$\lim_{t\to\infty} E_0^2\left[\|D^{(t)}\|_F\right] = 0$$

and thus also

$$\lim_{t\to\infty} E_0\left[\|D^{(t)}\|_F\right] = 0.$$

By Markov's inequality, the preceding panel implies that $\|D^{(t)}\|_F$ converges to 0 in probability.

Here we see that we get convergence even when we are still conditioned on the initial mixture-weight values. That is, we get convergence regardless of the starting mixture-weight values, other than they should be strictly between 0 and 1. Hence, we do not need any distributional assumptions on these starting weights $\Pi_i^{(t=0)}$.

B Example Posts Generated by Three Models at t=1 and t=60

- At $t = 1$, `Llama-3.1-8B-Instruct`: *Bitcoin's price is expected to remain stable around $30,000 as the market awaits the Federal Reserve's announcement on interest rates despite recent fluctuations in global hash rate and market volatility.*
- At $t = 1$, `deepseek-llm-7b-chat`: *Bitcoin's future prospects are expected to continue gaining mainstream adoption, with major institutions like PayPal, JPMorgan, and Tesla integrating crypto into their services and products.*
- At $t = 1$, `Mistral-7B-Instruct-v0.3`: *The future prospects of Bitcoin continue to evolve, with increasing institutional adoption, regulatory decisions, and the emergence of new stablecoins and CBDCs impacting its growth potential.*
- At $t = 60$, `Llama-3.1-8B-Instruct`: *The future prospects of Bitcoin are marked by growing mainstream adoption, institutional integration, increased user base, and regulatory decisions, while facing market volatility and security concerns.*
- At $t = 60$, `deepseek-llm-7b-chat`: *The future prospects of Bitcoin indicate continued mainstream adoption, influenced by increasing institutional integration, a growing user base, and regulatory decisions, while facing market volatility and security concerns.*
- At $t = 60$, `Mistral-7B-Instruct-v0.3`: *The future prospects of Bitcoin suggest continued mainstream adoption, driven by increasing institutional integration, a growing user base, and regulatory decisions, while facing market volatility and security concerns.*